# List of Topics

After the Midterm:

- Cylindrical Electrode Systems
- Electrical Breakdown in Gases
- Breakdown in Liquids
- Breakdown in Solids
- Corona Discharge
- Refraction of Electric Fields
- Approximated Electric Field Calculations
- Finite Difference Method
- Group Project Presentations on HV Test and HV Laboratories
- Academic and Industrial Invited Expert Speaker Lesson

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# Study Questions:

## Invited Expert Speaker Lesson

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1. Could you briefly explain ITU Fuat Kulun High Voltage and your major activites in that facility?
2. Among the subfields and topics of profession in High Voltage, what do you speficially focus on?
3. How would you decribe the field of High Voltage, compared to the other professions in Electrical and Electronics Engineering? (in terms of difficulty, required background, career opportunities, technological developments and other)
4. What are the career opportunities in the area of High Voltage? How would you compare it with other areas of Electrical and Electronics Engineering?
5. What are the working conditions and provided availabilities in the area of High Voltage? How would you compare it with other areas of Electrical and Electronics Engineering? (salary, working environment, future career pathways and other)
6. Which skills are required in High Voltage industry and academia? (softwares, mathematical and physical background, practice, hardware and hardware in the loop knowledge and other)
7. What are the major public and private organizations about High Voltage in Turkey?
8. What are the major public and private organizations about High Voltage in Europe?
9. What are the major public and private organizations about High Voltage in the World?
10. What are the major universities about High Voltage in Turkey?
11. What are the major universities about High Voltage in Europe?
12. What are the major universities about High Voltage in the World?
13. What are the current and near future challenges, issues, problems about High Voltage in the industry and academia?
14. What are the current and near future trends about High Voltage in the industry and academia?
15. How are the academic opportunities about High Voltage, compared to other areas of Electrical and Electronics Engineering?
16. What are the difficulties, challenges or disadvantages of the area of High Voltage, in industry and academia?

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# Study Questions:

## Cylindrical Electrode Systems

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1. Which equipment and components in HV systems can be considered as cylindrical electrode?
2. What is a coaxial cylindrical electrode system? Draw an example.
3. In the voltage formula, the minimum electric field strength formula and the maximum electric field strength formula for coaxial cylindrical electrode systems, which symbol represents which parameter in the real system?
4. Draw equipotential lines and field lines in an example coaxial cylindrical electrode system.
5. How a multilayer coaxial cylindrical electrode system can be represented, using capacitors?
6. Consider a multilayer concentric spherical electrode system numerical example we solved in previous weeks as a coaxial cylendrical electrode system. Using the same given numerical values, but using the formulas for cylindrical electrode systems, please solve the example.
7. How uniform stress condition can be ensured in multilayer coaxial cylindrical electrode systems?
8. In actual electrode seperation formula, effective electrode seperation formula, geometric characteristic formula and utilization factor (electrode efficiency formula), which symbol represents which parameter in the field?

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# Study Questions:

## Electrical Breakdown in Gases

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1 Explain Townsend Theorem in details. Also explain its two coefficients. Draw figure if needed.

2 Explain streamer theory. Draw figure if needed.

3Show electrode gap-breakdown voltage for three different electrode geometry and polarities on a single graph.

4 Explain breakdown, flashover and partial discharge. What is the difference between each?

5 Explain ionization by collision in details.

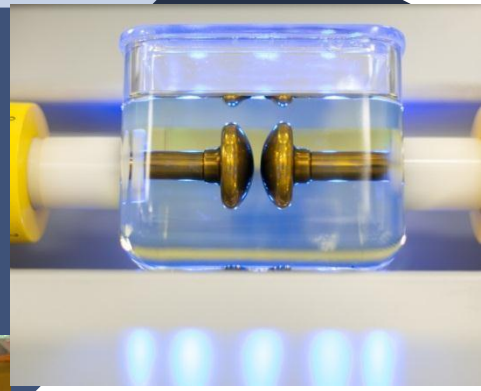
6 Explain photo-ionization in details.

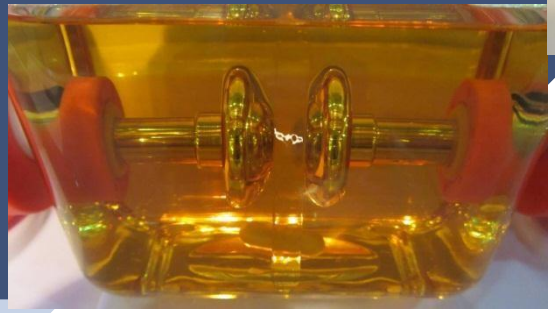
7 Explain thermal ionization in details.

8 Explain surface ionization in details, with its four sub categories.

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# Study Questions for:
## Breakdown in Liquids





Asst. Prof. Dr. M. Alparslan Zehir

# Study Questions

1. What are the three commonly used liquid dielectrics?
2. What are the three main reasons for using liquid dielectrics?
3. Write the 8 factors that affect breakdown strength and breakdown voltage in liquids. Briefly describe how they affect breakdown strength and breakdown voltage?
4. Explain the three types of impurities and how they behave in liquid dielectrics.
5. Write the three types of breakdown in liquid dielectrics and briefly describe each.
6. How dissolved gasses , water globules and solid particles behave in liquid dielectrics? Describe each seperately.
7. Write the desired dielectric oil properties.
8. On a $U_{breakdown}/electrodegap$ graph, show the impact of electrode polarity and geometry on breakdown voltage.

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# Study Questions:

## Breakdown in Solids

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# STUDY Questions

1 List three solid dielectric materials.

2 Write the names of four losses in solid dielectrics.

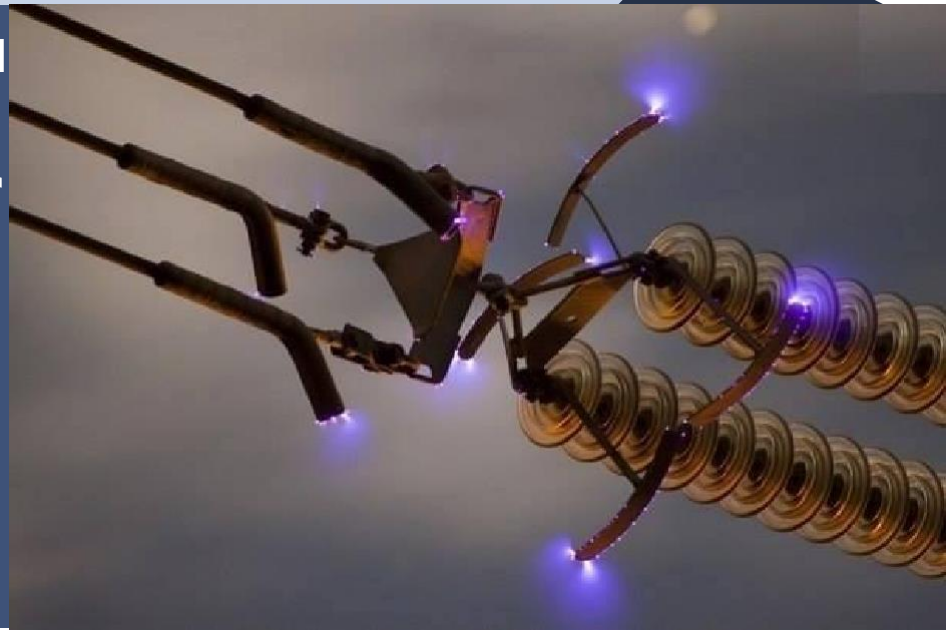
3 List three breakdown types in solid dielectrics.

4What is electromechanical breakdown? How it happens?

5- What is thermal breakdown? How it happens?

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# Study Questions for
## Corona Discharge



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# Study Questions for Corona Discharge

1. What is the literal meaning of corona?

2.What is the physical meaning of corona in HV? Which places-parts of equipment does it occur?

3.What are the three main indicators of corona discharges? What are the impacts of these indicators?

4. How weather conditions impact corona?

5. In the corona voltage formula, please describe which symbol represents which parameter in the real world.

6. In the power loss formula (Peek's formula) please describe which symbol represents which parameter in the real world.

7. How the corona discharges can be minimized?

8. Explain the ozone gas generation related problems of corona discharges.

9. Explain the corona discharge problems in $SF_6$ gas insulated systems.

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# Study Questions:

## Refraction of Electric Fields

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1 In which cases we focus on refraction of electric fields?

2 Show the refraction of electric field lines and equipotential lines on an example case, labelling the angles, and the vectors.

3Solve the two examples again, using different parameters, modifying the scenarios given in the question.

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# Study Questions:

## Approximated Electric Field Calculations

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1 A plane-sphere electrode system can be approximated to which system? How the original equation is modified?

2A sphere-sphere electrode system can be approximated to which system? How the original equation is modified?

3Solve the sphere-sphere numerical example again, using different parameters, modifying the scenario given in the question.

4A plane-cylinder electrode system can be approximated to which system? How the original equation is modified?

5 A cylinder-cylinder electrode system can be approximated to which system? How the original equation is modified?

6 Solve the cylinder-cylinder numerical example again, using different parameters, modifying the scenario given in the question.

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# Study Questions:

## Finite Difference Method

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# STUDY Questions

1 What is the general formula for calculating the voltage of a selected point, using finite difference method.

2 Solve the two examples again, using different parameters, modifying the scenarios given in the question.

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# Study Questions:

## Group Projects on HV Test and HV Laboratories

Asst. Prof. Dr. M. Alparslan Zehir

# STUDY Questions

1 List the names, cities and countries of three HV laboratories presented by the groups.

2 List the names of three tests presented by the groups. Explain a test presented by another group in details (not your own group’s test).

# END OF THE LESSON

Any questions?